## Simulation Parameters

Actual vehicle speed (km/h)

45

Radar angle θ (degrees)

30

Signal-to-noise ratio (dB)

10

Noise random seed

0

## Warning thresholds

Green/Yellow threshold (km/h)

50

Yellow/Orange threshold (km/h)

# Doppler Radar Vehicle Speed Estimation Using DSP

CW Doppler radar simulation: signal generation → noise → FIR filtering → FFT peak detection → radial-speed estimation → angle correction.

## Estimated Results

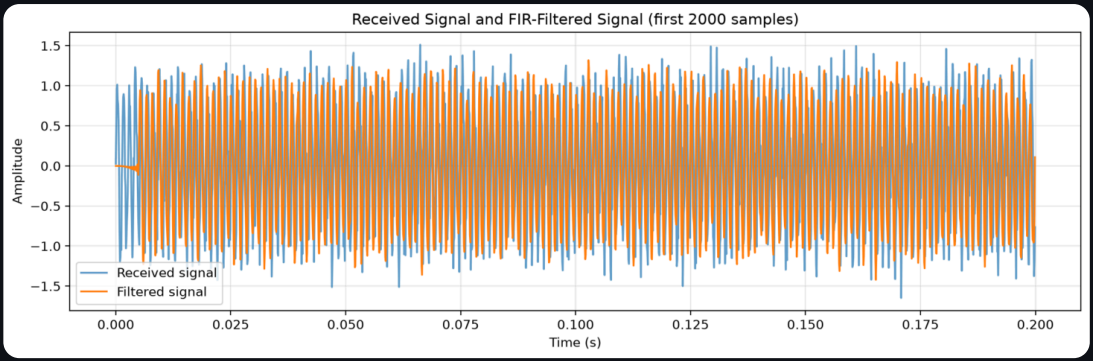
| Actual speed | Radial speed (actual) | Estimated speed (c… | Doppler frequency… |
|---|---|---|---|
| 45.00 … | 38.97 … | 44.98 … | 759.6 … |

**Speed Warning: GREEN — SAFE**

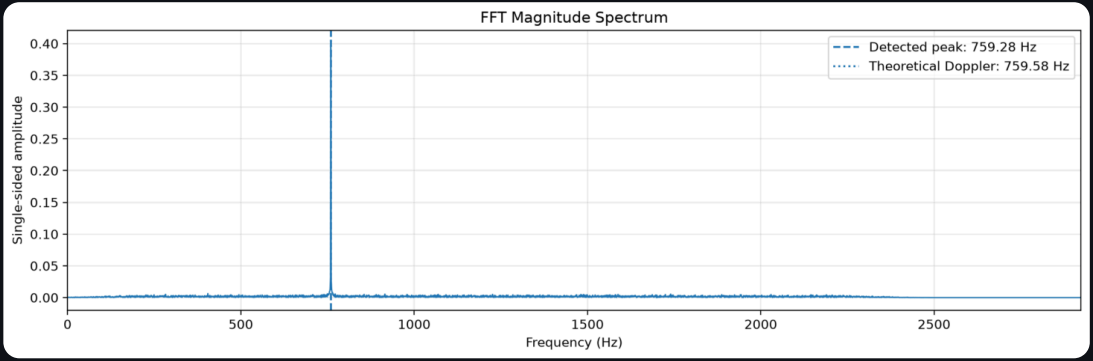
**Estimated Doppler frequency:** 759.28 Hz | **Absolute error:** 0.02 km/h | **Percentage error:** 0.04%

Angle correction uses the selected angle: $v = v_r / \cos(\theta)$. A single-channel CW Doppler signal does not independently determine the angle. This is a synthetic educational simulation, not measured radar data.

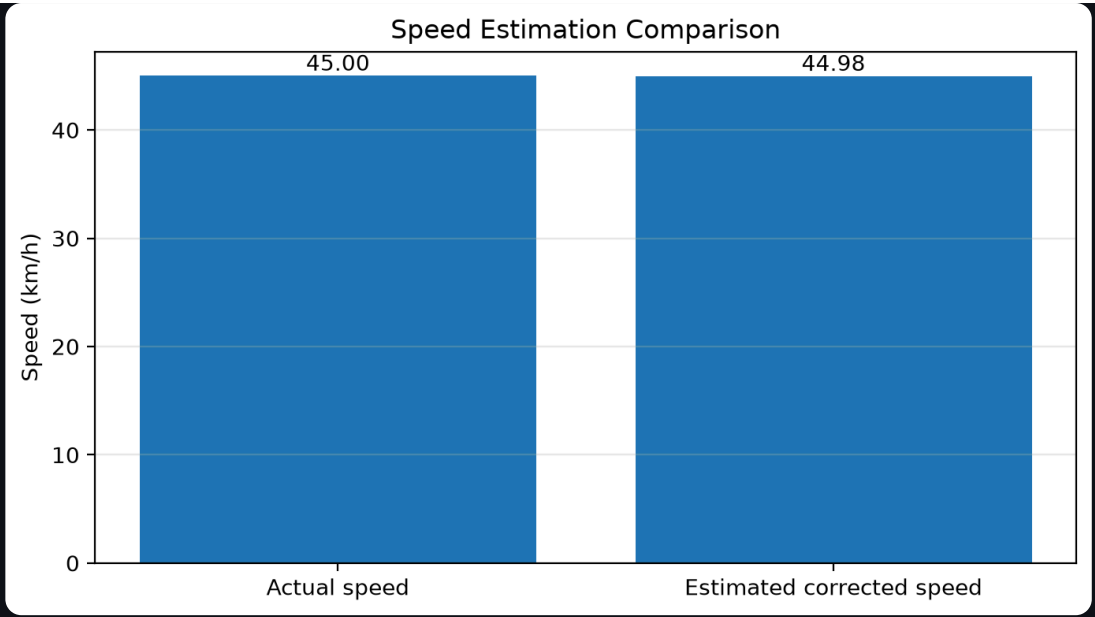
# 1. Time-Domain Signals



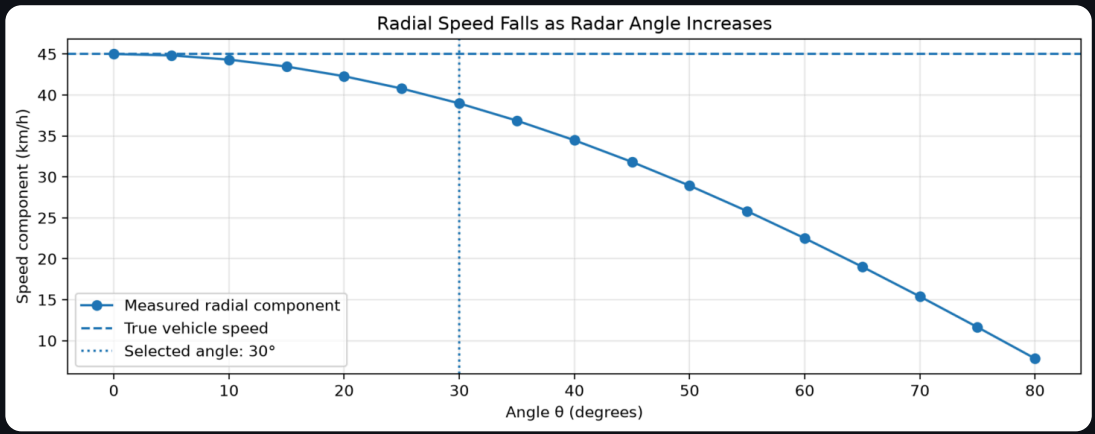

# 2. FFT Spectrum and Detected Doppler Peak



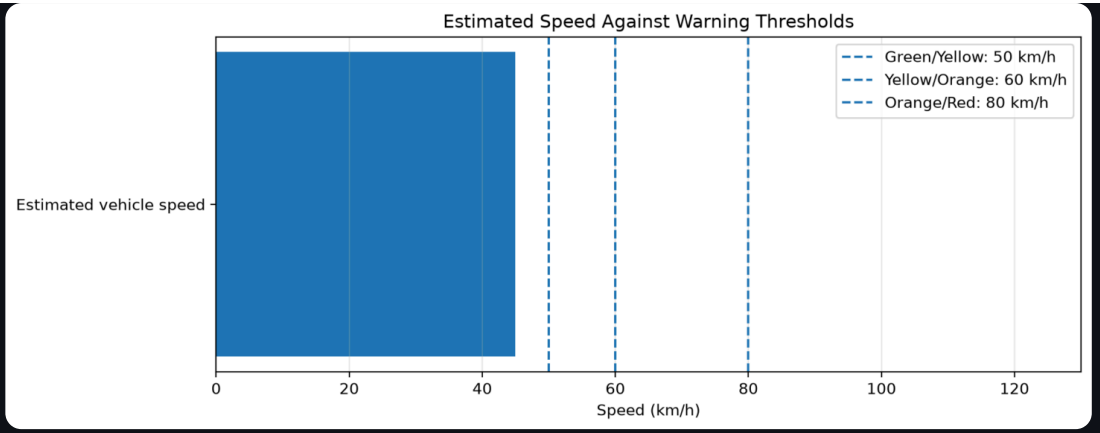

# 3. Actual vs Estimated Speed




## 4. Effect of Radar Angle




## 5. Speed Warning Levels




Show simulation parameters and DSP details


```
{
  "Radar carrier frequency (Hz)" : 10525000000
  "Speed of light (m/s)" : 300000000
  "Wavelength (m)" : 0.028503562945368172
  "Sampling frequency (Hz)" : 10000
  "Simulation duration (s)" : 2
  "Number of samples" : 20000
  "FFT length" : 16384
  "SNR (dB)" : 10
  "FIR filter order" : 100
  "Minimum Doppler frequency (Hz)" : 97.4537037037037
  "Maximum Doppler frequency (Hz)" : 2338.888888888889
  "Theoretical Doppler frequency (Hz)" :
  759.5764479026014
  "Detected Doppler frequency (Hz)" : 759.27734375
```

```
    "Angle correction used" : "v = v_r / cos(30°)"
}
```

Note: The FIR filter uses a Hamming-window design. FFT-bin spacing and noise can cause small differences between actual and estimated speed. Warning thresholds are demonstration values.